The Year of Yes! How Saying “Yes” Ignited GSK’s R Adoption

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PHUSE US Connect 2026

YES

Director of Data Science

Scientific and Adoption Lead for Open Source for GSK R&D

Passionate about open source and teaching

Associate Director of Data Science

Enablement Lead for Open Source for GSK R&D

Passionate about user-focus and community


$2,400
$6,800
$4,200

INTRO	FIRST 3 MONTHS	SCALING UP	LOOKING FORWARD	FINAL JEOPARDY
$200	$200	$200	$200	$200
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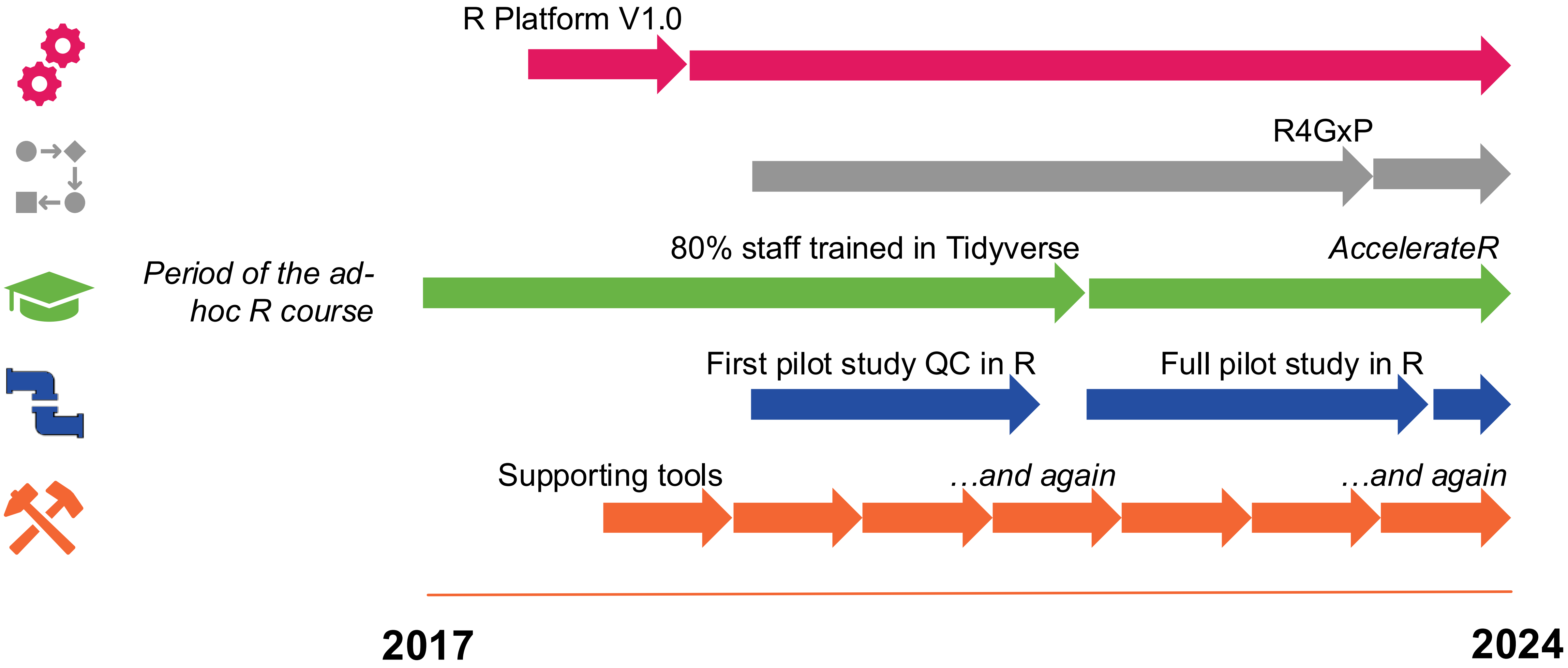
CAN WE DO 50% OF OUR PROGRAMMING IN R?

GSK's Open-source Journey Thus Far



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GSK Biostatistics’ Commitment to Open-Source

All central tools to be built using open-source languages

50% of our code to be open-source by end 2025

Photo by Arturo Añez on Unsplash

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GSK Biostatistics’ Commitment to Open-Source

All central tools to be built using open-source languages

70%

~~50%~~ of our code to be open-source by end 2025

stretch goal

Photo by Arturo Añez on Unsplash

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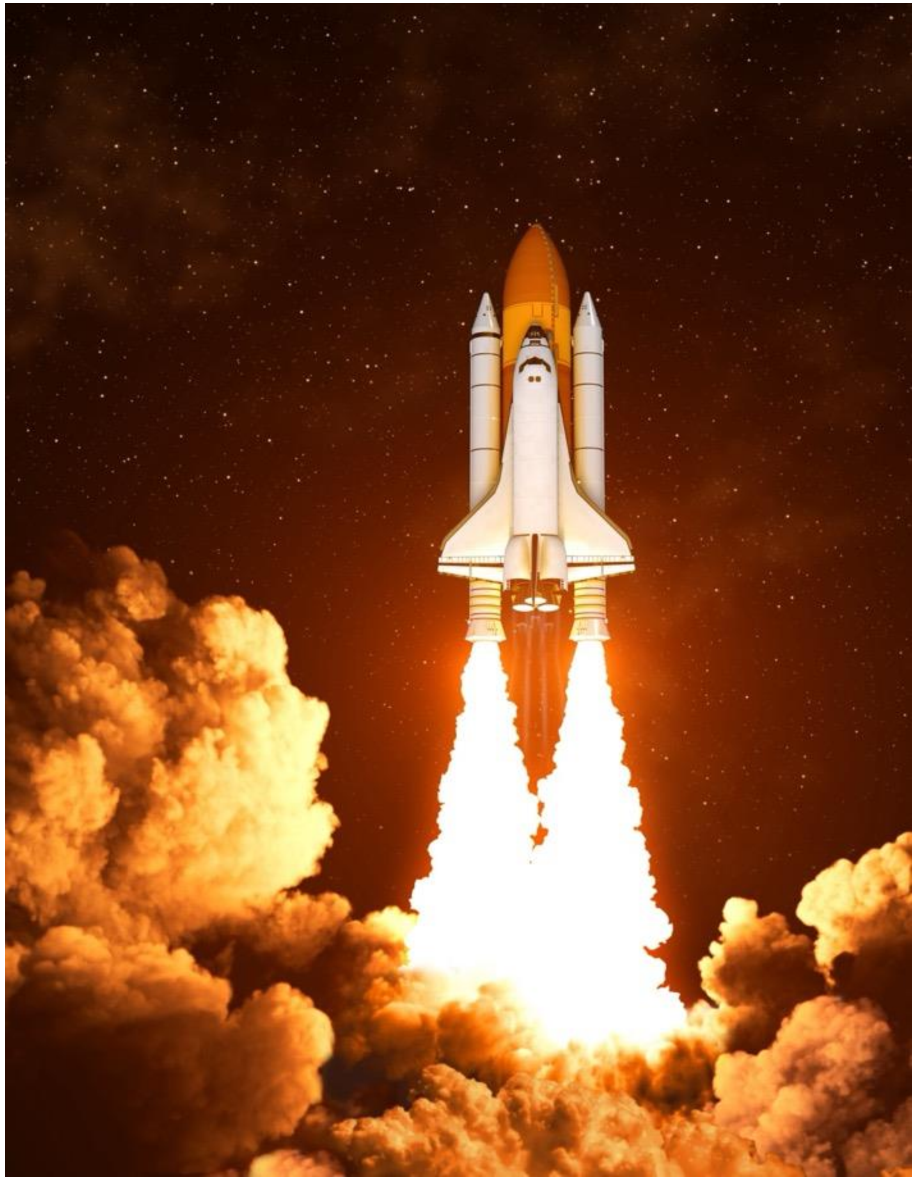
STAR

Starting in 2025: GSK’s R&D-Wide Open-Source Adoption Program

Photo by Chantal & Ole on Unsplash

Kicking off with a “burst” of support

GSK Jointly launched STAR and **STARburst, a support team comprised of R experts**



Aims:

- Support STAR and the Biostatistics R Adoption Objective
- Enable skills development in the pipeline via *real-time* mentoring / support model
- Identify skills, tools, resource gaps

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CAN WE USE R END-TO-END?

STARburst: 3-months of “yes” to every inquiry

“Dear STARburst team…”

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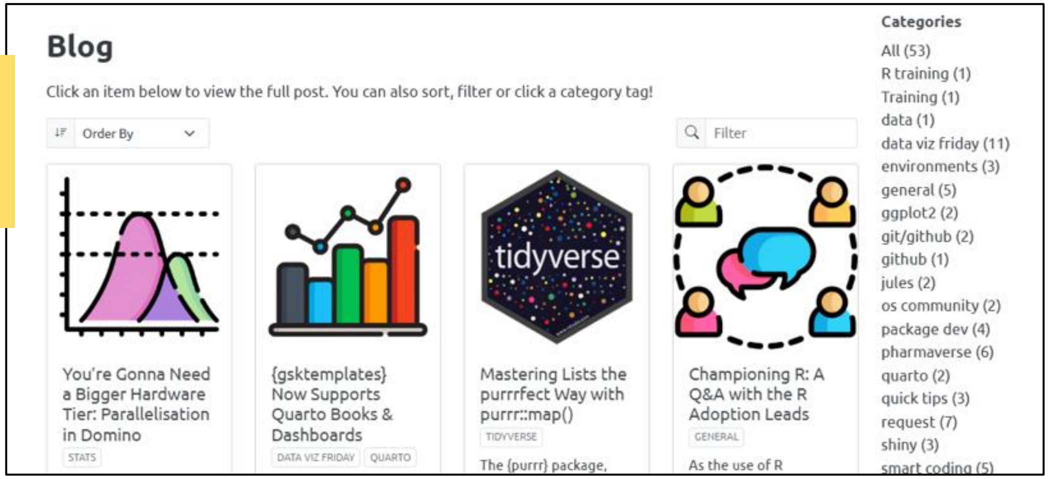
Unblocking and Upskilling

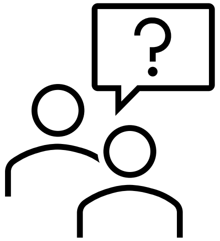
```
+    geom_boxplot()
Error: object 'ADSL' not found
```

Tackling requests with an open-source mentality

Internal sharing

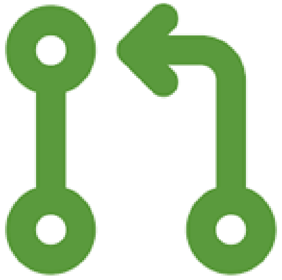


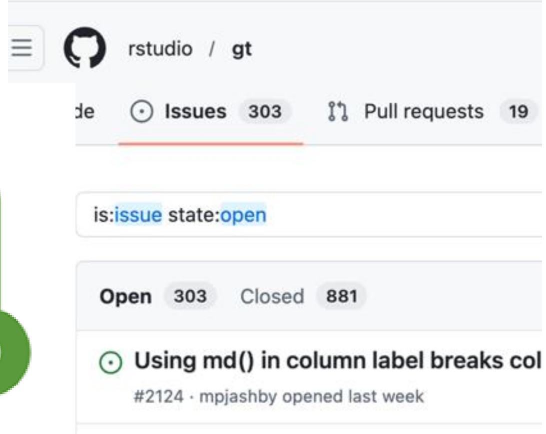
STARburst Team



Study Team

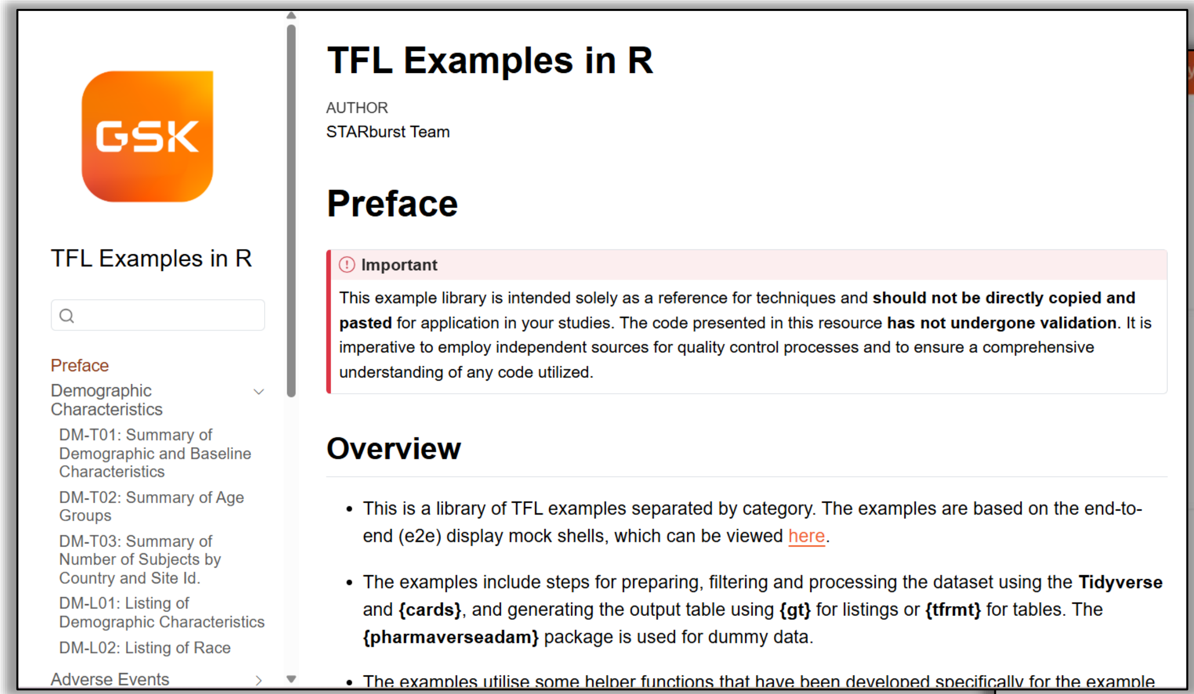
External Sharing



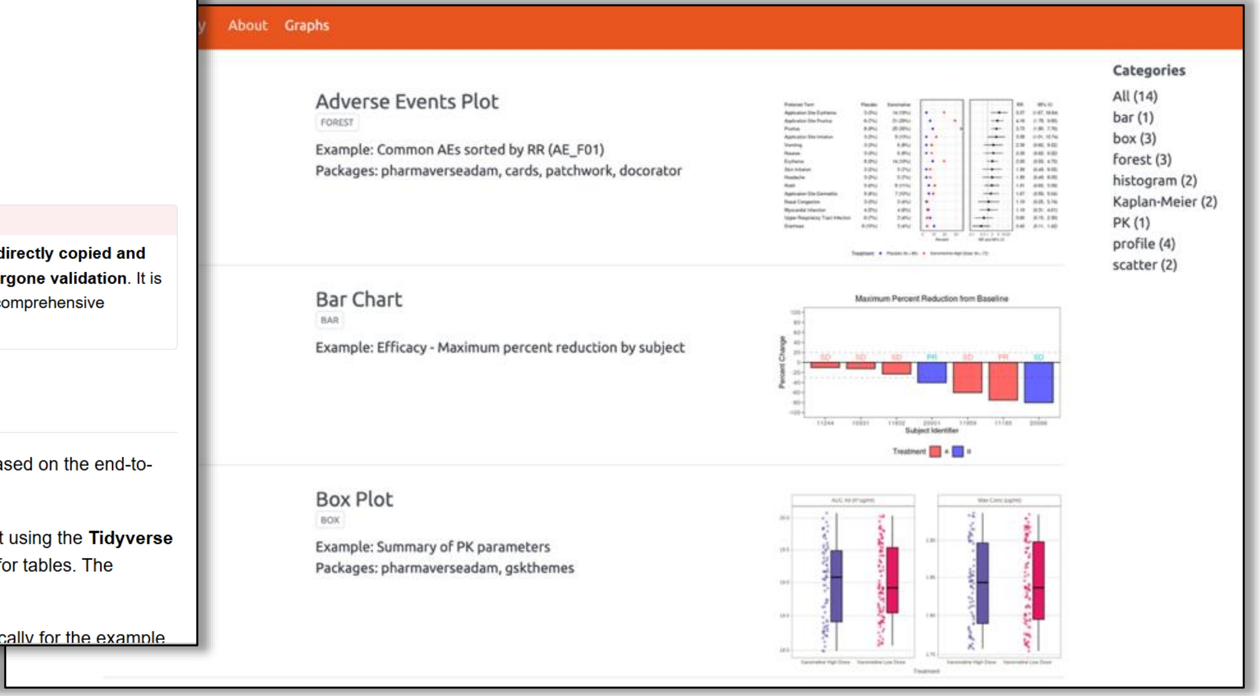


Applying our learnings in real time

Library of standard TFLs



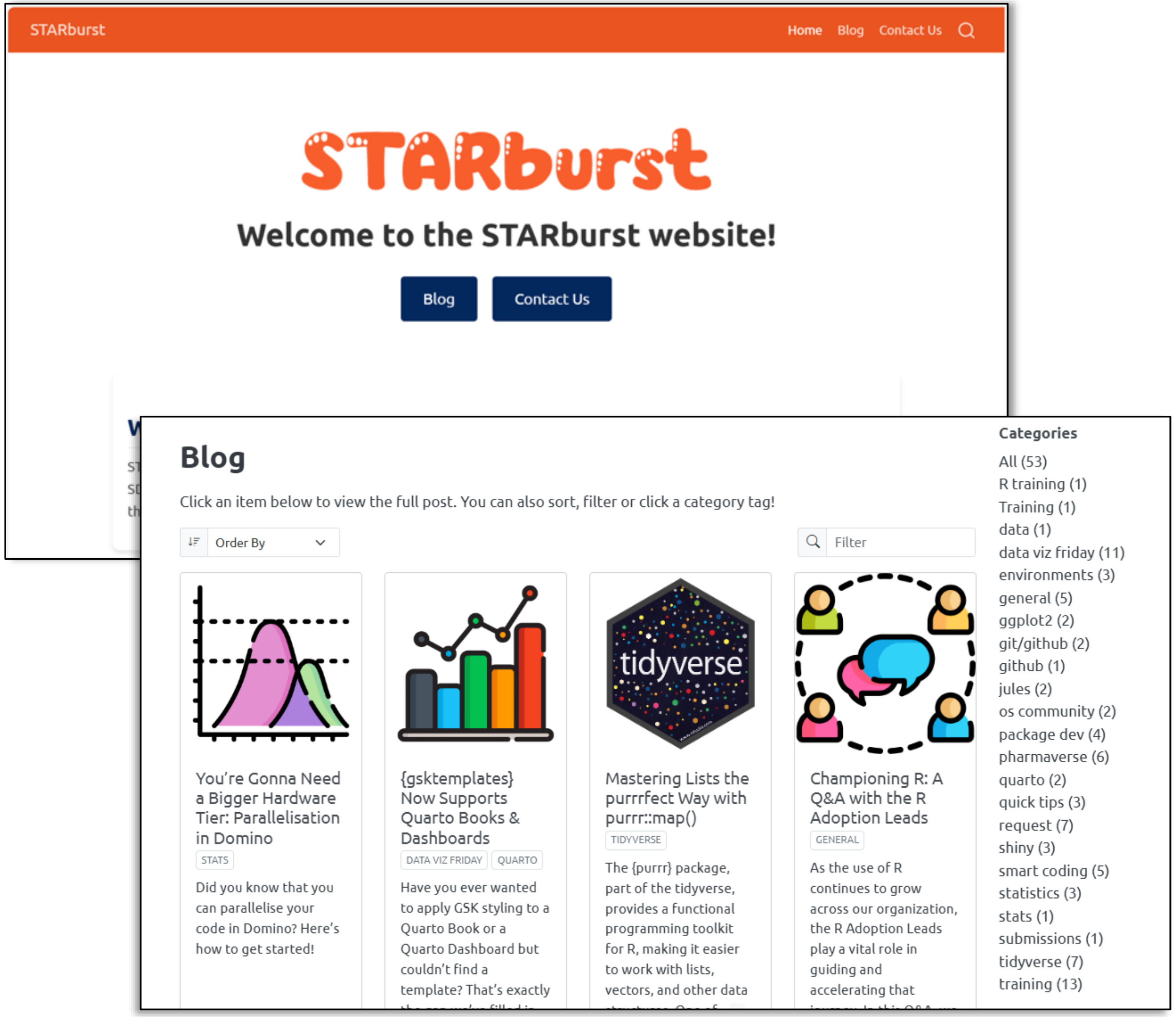
Library of visualization techniques



STARburst by the numbers

A look at the first 3 months

- **64 blog posts (posted daily)**
  - Topics driven by the pipeline, featuring case studies and tricks/tips
  - 821 unique viewers (~90% returning)
  - 16004+ views
- **58 requests (~1 per day)**
  - Programming - 47%
  - Statistics - 48%



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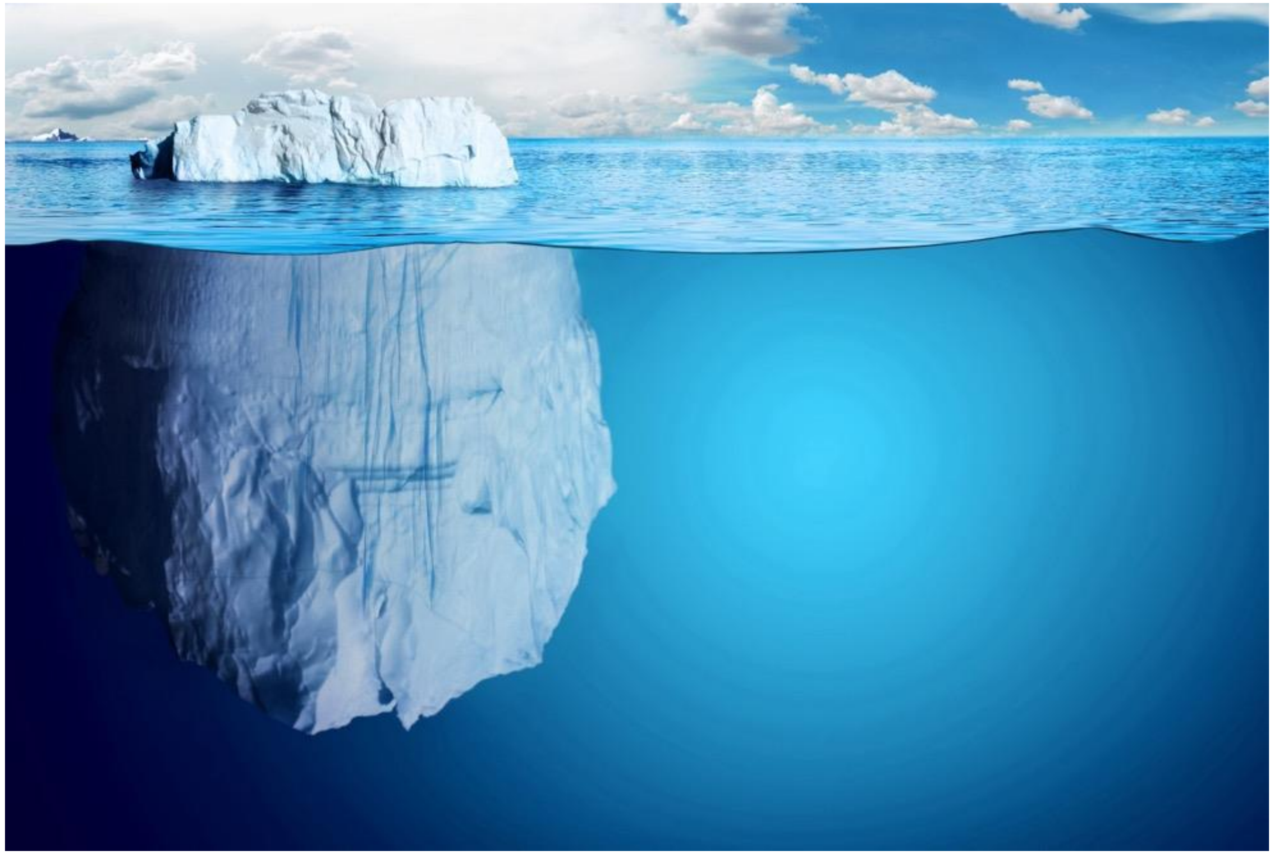
DAILY
DOUBLE

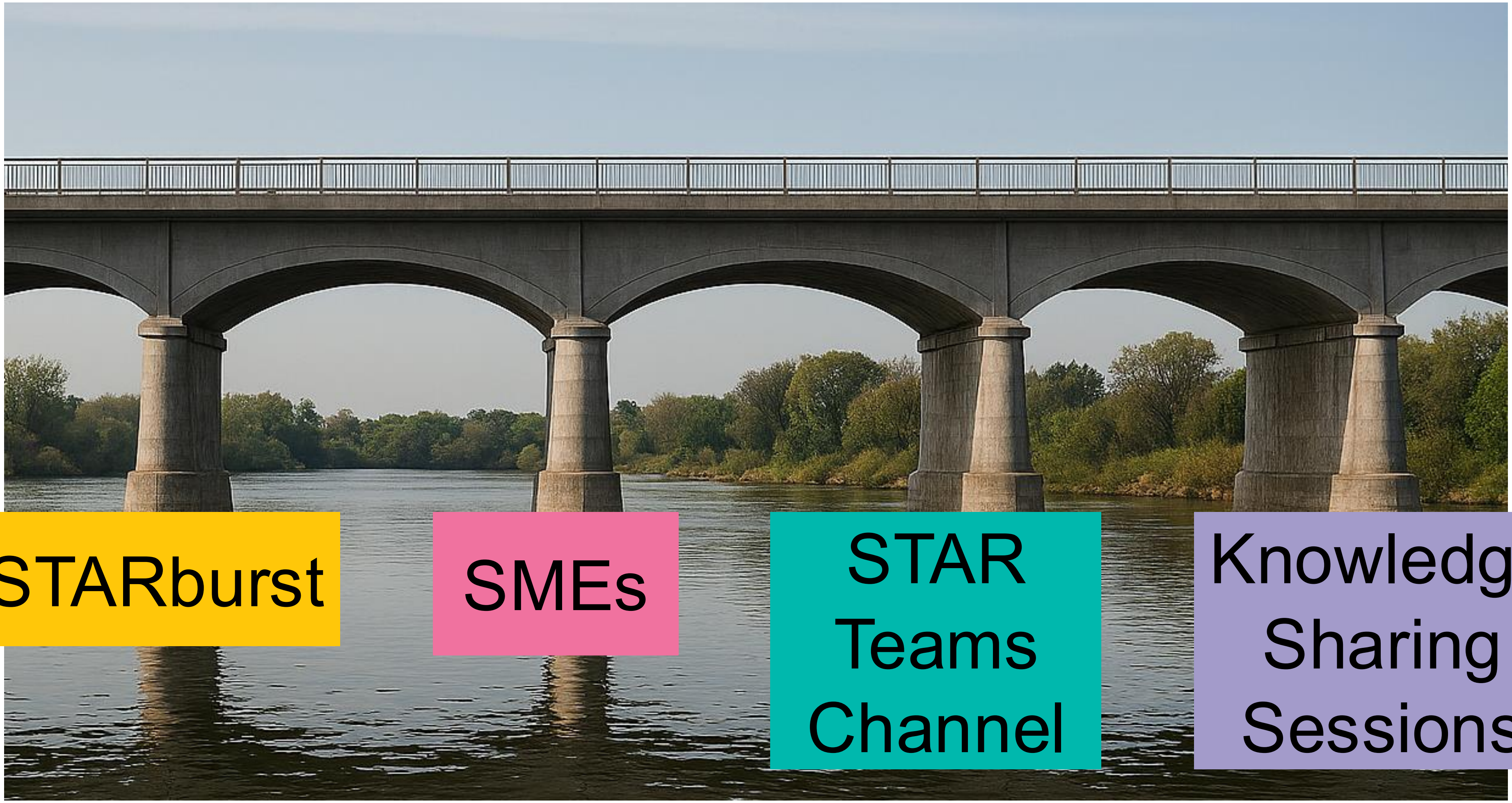
CAN WE TURN OUR LEARNINGS INTO SHARED PRACTICE?

Celebrating our impact, recognizing the challenge ahead

- Users want more than isolated examples, they want step-by-step workflows
- Each department, therapeutic area, study, regulatory agency, etc. presents unique scenarios
- We don’t know what we don’t know



Scaling through community-wide support



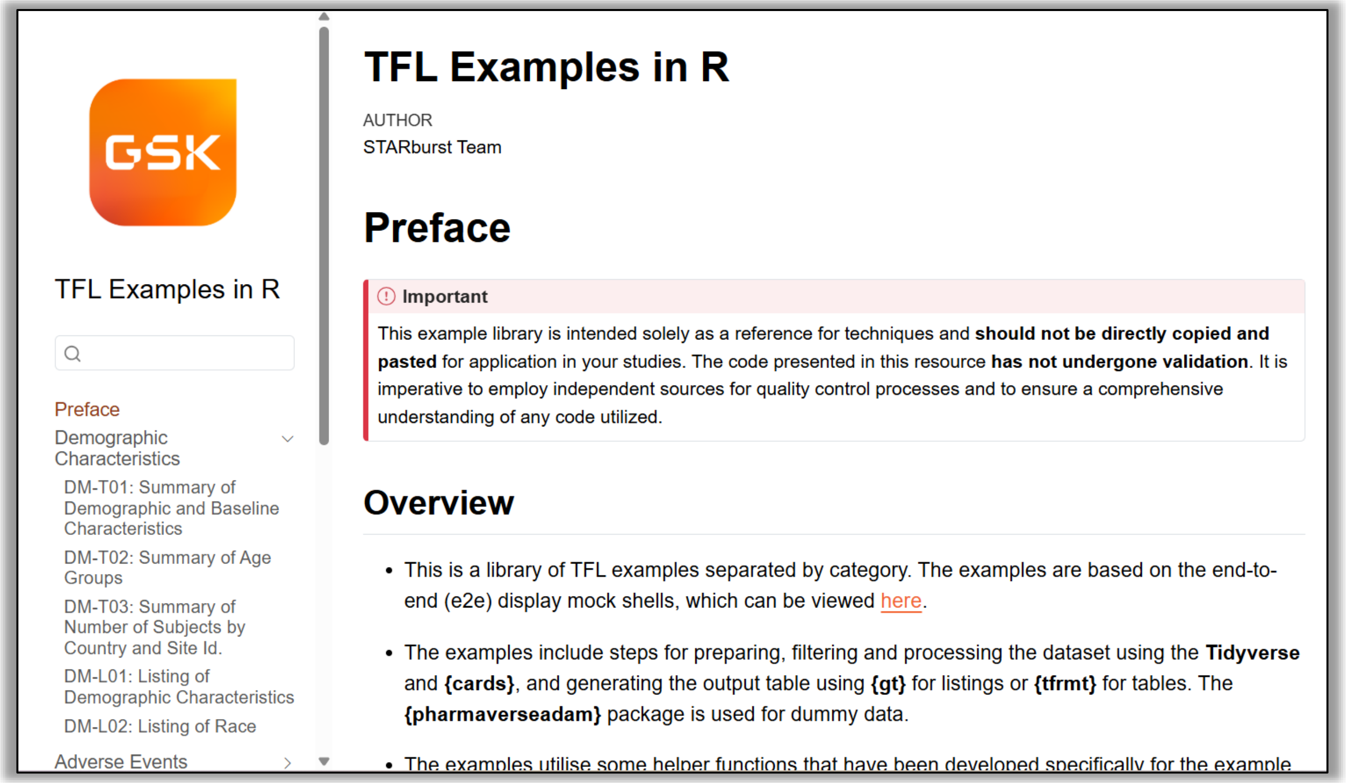
STARburst

SMEs

STAR Teams Channel

Knowledge Sharing Sessions

Scaling through a wider variety of examples



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TFL Examples in R

Preface

Demographic Characteristics

DM-T01: Summary of Demographic and Baseline Characteristics

DM-T02: Summary of Age Groups

DM-T03: Summary of Number of Subjects by Country and Site Id.

DM-L01: Listing of Demographic Characteristics

DM-L02: Listing of Race

Adverse Events

TFL Examples in R

AUTHOR

STARburst Team

Preface

Important

This example library is intended solely as a reference for techniques and **should not be directly copied and pasted** for application in your studies. The code presented in this resource **has not undergone validation**. It is imperative to employ independent sources for quality control processes and to ensure a comprehensive understanding of any code utilized.

Overview

- This is a library of TFL examples separated by category. The examples are based on the end-to-end (e2e) display mock shells, which can be viewed here.
- The examples include steps for preparing, filtering and processing the dataset using the **Tidyverse** and **{cards}**, and generating the output table using **{gt}** for listings or **{tfrmt}** for tables. The **{pharmaverseadam}** package is used for dummy data.
- The examples utilise some helper functions that have been developed specifically for the example

+ FAQs

CAN WE FIGURE OUT WHAT WE DON'T KNOW?

Scaling through robust workflows

"Tiger Team" assembled to deep dive on and share detailed workflows

Photo by Roger Starnes Sr on Unsplash

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Scaling through interdisciplinary problem solving

- Applied the “Tiger Team” concept to targeted challenges
- Brought together different voices
- Shifted focus from quick fixes to long-term solutions

Photo by Helena Lopes on Unsplash

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CAN WE CONTINUE TO GROW IN A SELF-SUSTAINING MANNER?

Requirements for self-sustaining R adoption

- Proactively identify and close gaps
- Deepen technical proficiencies
- Embrace continuous learning and adaptation
- Foster an open-source culture

The ultimate goal

Working toward a future where **innovation is shared** and **collaboratively maintained**

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FINAL JEOPARDY

CAN WE DO 50% OF OUR PROGRAMMING IN R?

CAN WE USE R END-TO-END?

CAN WE TURN OUR LEARNINGS INTO SHARED PRACTICE?

CAN WE FIGURE OUT WHAT WE DON'T KNOW?

CAN WE CONTINUE TO GROW IN A SELF-SUSTAINING MANNER?

What is Yes!

Questions?

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